RAJALAKSHMI ENGINEERING COLLEGE
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DEPARTMENT OF CSE

LEXVIGIL

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AI LEGAL INTELLIGENCE

Evidence-linked contract review prototype

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EXECUTIVE ABSTRACT

- LexVigil is a contract-review workspace that extracts supported documents and preserves clause-level source context.
- It identifies numbered clauses, definitions, explicit references, and amendments before running selected deterministic checks.
- The reviewer sees a candidate finding with its evidence trail and retains responsibility for interpretation.
- A synthetic services agreement shows an amended payment period compared with a related late-payment trigger.
- The current build is a development prototype. It does not provide legal advice or claim legal accuracy.

PROBLEM STATEMENT

Contract review becomes difficult when one obligation depends on clauses spread across a long document.

Long documents

Dates, duties, and exceptions appear in separate places.

Linked provisions

Definitions, references, and amendments need joint review.

Trustworthy review

A reviewer needs the source passage behind each flag.

How can a system help reviewers locate potential inconsistencies while preserving source context and human judgment?

LITERATURE SURVEY

CUAD

Expert-annotated contract review dataset covering 41 clause categories.

ContractNLI

Document-level legal inference with supporting evidence spans.

LegalBench

Broad benchmark for measuring legal reasoning tasks.

Implication for LexVigil

Contract analysis needs source evidence, long-document context, and task-specific evaluation.

RESEARCH GAP AND OBJECTIVES

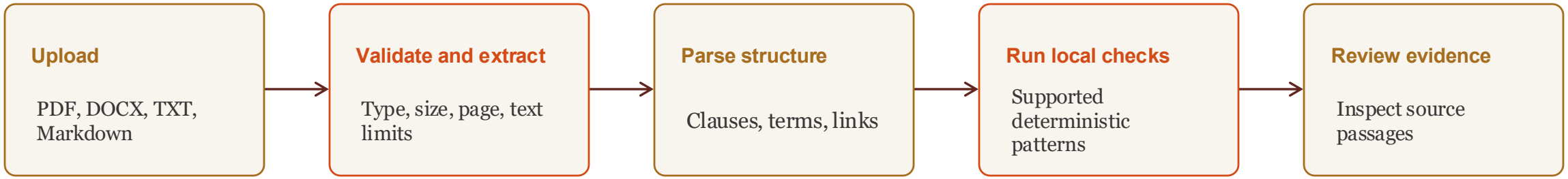
Research gap

Individual clause classification does not by itself connect definitions, cross-references, and amendments across a long agreement.

Objectives

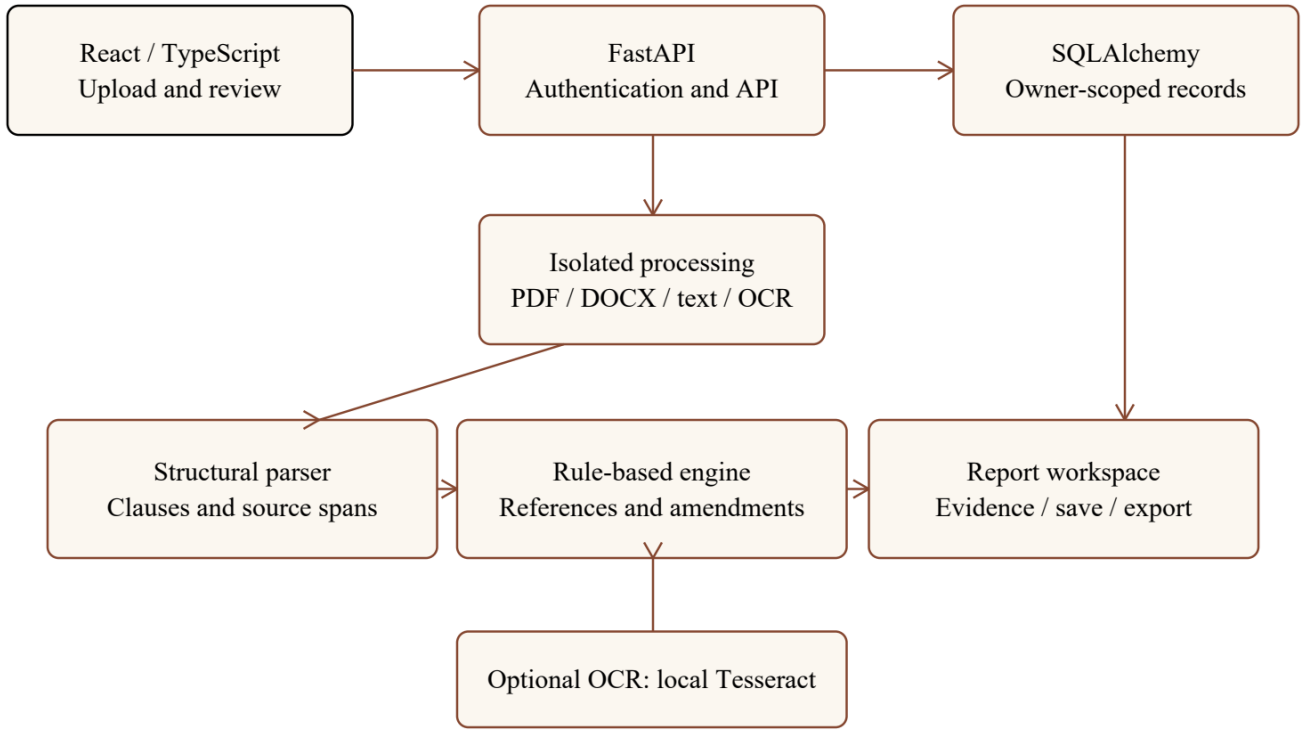
- Extract supported documents with provenance.
- Map clauses, terms, references, and amendments.
- Surface selected candidate issues with evidence.
- Keep review decisions with the human reviewer.

PROPOSED WORK



A candidate finding always carries a path back to the document evidence.

SYSTEM ARCHITECTURE



Live AI research and semantic review are currently unavailable.

Live AI research and semantic review are currently unavailable in the reviewed configuration.

ALGORITHM

1. Validate file and processing limits
2. Extract text and source positions
3. Use local OCR when eligible
4. Divide content into clauses
5. Resolve terms, references, and amendments
6. Apply supported deterministic checks
7. Attach evidence spans
8. Return candidates for human review

Output: a traceable candidate issue, not an automated legal opinion

IMPLEMENTATION

Technology stack

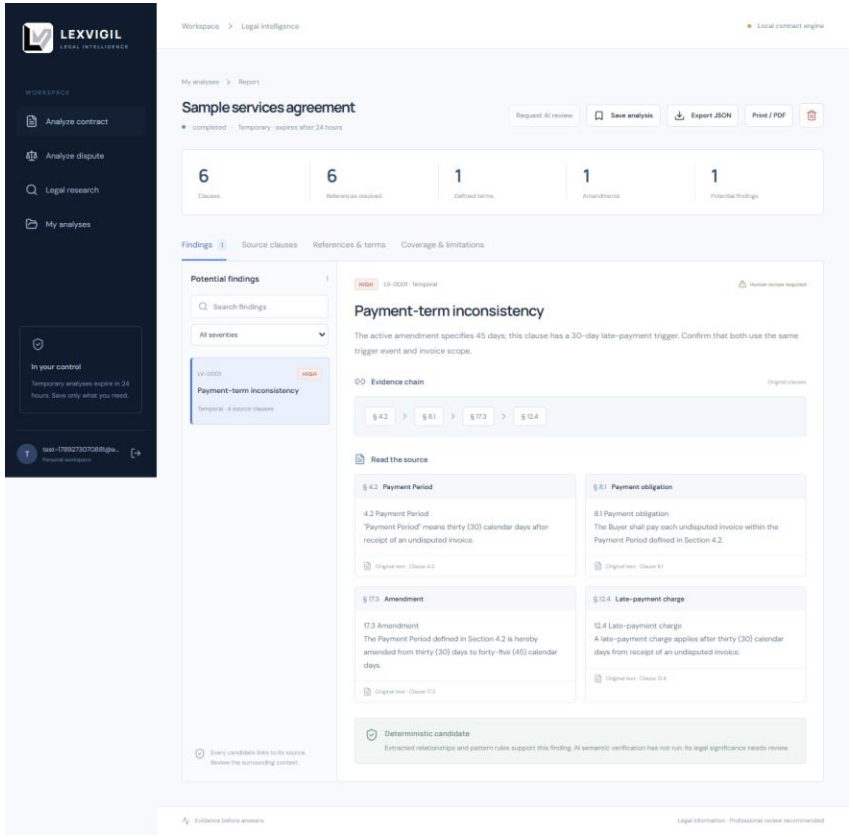
- Frontend: React 19, TypeScript, React Router, Vite
- Backend: FastAPI, Pydantic, SQLAlchemy
- Document parsing: PyMuPDF and python-docx
- OCR: optional local Tesseract for English, Tamil, and Hindi

Operational controls

- 15 MB files and 200-page document limit
- Temporary unsaved reports expire after 24 hours
- Consent-based original retention up to 48 hours in development
- Bounded worker resources, deadlines, and recovery attempts

Current status: development prototype, not production-approved

FINAL RESULTS AND OUTPUTS

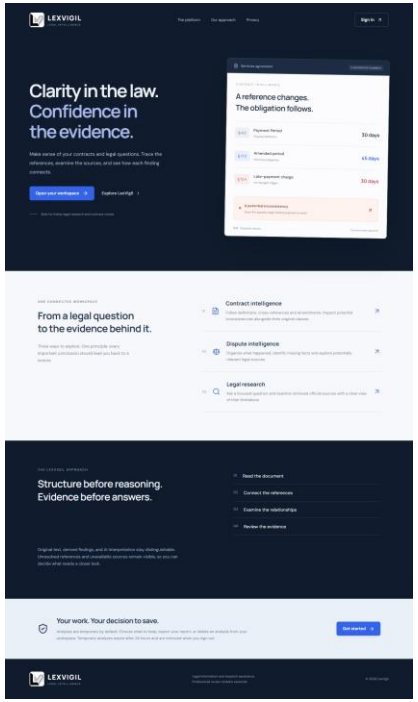


Synthetic services-agreement demonstration

- 6 clauses parsed
- 6 references resolved
- 1 defined term
- 1 amendment
- 1 potential finding

The payment term changes from 30 to 45 days, while a related late-payment trigger remains at 30 days.

OUTPUT INTERPRETATION



What the output provides

- A specific candidate finding
- Clause identifiers and linked evidence
- Original passages for source inspection
- A clear reminder that human review is required

The interface supports inspection. It does not decide the legal meaning of a contract.

VALIDATION AND LIMITATIONS

Evidence from the reviewed development session

13 focused tests

Selected parser, amendment, reference, and ingestion behaviours passed.

No legal benchmark

No expert-labelled Indian contract evaluation has been completed.

Bounded coverage

Full taxonomy coverage and live AI semantic review remain unavailable.

The tests verify selected software behaviour. They do not establish legal correctness or completeness.

CONCLUSION AND FUTURE WORK

LexVigil connects selected contract-review candidates to the passages that support them.

Current contribution

- Structured clause relationships
- Evidence-linked candidate findings
- Human review remains central

Next steps

- Build an expert-labelled Indian contract dataset
- Measure accuracy and reviewer time
- Expand coverage, security review, and deployment readiness

LexVigil supports contract review; legal professionals remain responsible for final interpretation.

REFERENCES

- Hendrycks, D., et al. (2021). CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review. NeurIPS Datasets and Benchmarks.
- Koreeda, Y., and Manning, C. D. (2021). ContractNLI: A Dataset for Document-Level Natural Language Inference for Contracts. Findings of EMNLP.
- Guha, N., et al. (2023). LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models. NeurIPS Datasets and Benchmarks.
- LexVigil project repository: implementation documentation, screenshots, test and release-scope records reviewed for this presentation.

Thank you